distant images would allow us (with a modest radio telescope) to resolve a continent at the distance of the nearest star and the inner Solar System at the distance of the nearest spiral galaxy. If you are free to roam an imaginary spherical shell at the appropriate focal distance and centered on the Sun, you are free to explore the Universe in stupendous magnification, to peer at it with unprecedented clarity, to eavesdrop on the radio signals of distant civilizations, if any, and to glimpse the earliest events in the history of the Universe. Alternatively, the lens could be used the other way, to amplify a very modest signal of ours so it could be heard over immense distances. There are reasons that draw us to hundreds and thousands of AU. Other civilizations will have their own regions of gravitational focusing, depending on the mass and radius of their star, some a little closer, some a little farther away than ours. Gravitational lensing may serve as a common inducement for civilizations to explore the regions just beyond the planetary parts of their solar systems.

Second: Spend a moment thinking about brown dwarfs, hypothetical very low temperature stars, considerably more massive than Jupiter, but considerably less massive than the Sun. Nobody knows if brown dwarfs exist. Some experts, using nearer stars as gravitational lenses to detect the presence of more distant ones, claim to have found evidence of brown dwarfs. From the tiny fraction of the whole sky that has so far been observed by this technique, an enormous number of brown dwarfs is inferred. Others disagree. In the 1950s, it was suggested by the astronomer Harlow Shapley of Harvard that brown dwarfs—he called them “Lilliputian stars”—were inhabited. He pictured their surfaces as warm as a June day in Cambridge, with lots of area. They would be stars that humans could survive on and explore.

Third: The physicists B. J. Carr and Stephen Hawking of Cambridge University have shown that fluctuations in the density of matter in the earliest stages of the Universe could have generated a wide variety of small black holes. Primordial black holes—if they exist—must decay by emitting radiation to space, a consequence of the laws of quantum mechanics. The less massive the black hole, the